

Project 1 Phase 0 – Problem Statement

MavDine – A Digital Cookbook Platform for UTA Students

UTA is planning to start “MavDine” – an online platform exclusively dedicated to cookbooks *by and for* UTA students. This platform serves as a comprehensive digital hub for culinary enthusiasts, offering a vast and diverse library of dynamic cookbooks from a wide variety of cuisines.

The system UTA is planning to design will keep track of all Home Cooks, Chefs, Cookbooks, Recipes, Merchants, Campaigns, as well as the relationships among them. This is an ideal solution for those seeking a dedicated platform that brings together several different student-oriented cookbooks. This service is crafted to meet the evolving food culture, making it an essential tool for anyone who loves to turn their kitchen into a flavor hotspot.

Data Description

Every MavDine user (that is a UTA student) has a unique maviD and will have other basic information such as **full name comprising of first name, and last name**, username, email, password, phone number, **date of birth**, date of enrollment, cumulative GPA, enrolled department, nationality, and their full physical address comprising of the street address, apartment#, county and zip code, **and city where they live**. These will be populated when they initially register on our platform. We will derive users **age** using date of birth. There are two types of users – home cooks and chefs. It is possible for a user to be both a home cook and a chef.

Home Cook

- We define a home cook as a UTA student who subscribes to cookbooks, reads recipes, follows/unfollows chefs and engages with the recipes through likes and dislikes.
- The home cooks will also have their reading related information stored to capture **log as composite multi-valued attribute, comprising start time and end time**, from which location and for how long they read a particular recipe is derived, **and how many times they read a recipe is derived from the number of logs recorded**.

Chef

- A chef is a student on our platform who creates new cookbooks. Moreover, a given cookbook is created by only one chef.
- Each chef has a unique chefID and an additional attribute that determines the number of followers they have. This number is different from the subscriber count of their individual cookbooks.

Recipe

- A recipe is the text file published by chefs for a particular cookbook they own. Each recipe has a recipe number, which is stored in a sequentially increasing manner starting from 1.
- After a new recipe is published, all the cookbook subscribers will get a notification.
- A recipe will have attributes such as title, source - the URL where the actual file is physically stored, date and time of publish, meal type, list of 3 main ingredients,

preparation time, cooking time, and total calories. Every recipe has a single meal type -

breakfast, brunch, lunch, snacks, dinner or dessert. Additionally, important stats like the number of reads, likes and dislikes for a recipe can be calculated.

- A recipe can only belong to a single cookbook. When the cookbook is deleted, all the recipes of that cookbook are also deleted.

Cookbook

- A cookbook is defined as a set of recipes published by chefs, to which home cooks can subscribe and get notifications when new ones are published.
- A chef can create multiple cookbooks and, of course, one cookbook can have multiple recipes.
- Each cookbook has a unique ID, name, date of creation, description, number of subscribers and a pre-defined primary cuisine associated with it. Currently, supported cuisines are Chinese, French, Greek, Indian, Italian, Japanese, Korean, Mediterranean, Mexican, Spanish, and Thai.
- A cookbook can only be for a single cuisine.

Merchant

- A merchant is a third-party grocery brand that is interested in advertising their campaigns on our platform.
- Each merchant is identified by a unique registration number, name, website, company headquarter city, and the number of employees.
- Merchants are responsible for creating the campaign and providing our platform with a hyperlink.
- A merchant can advertise multiple campaigns. If the merchant is deleted, then all the campaigns belonging to that merchant are also deleted.

Campaign

- We define a campaign as the promoted content of a registered merchant that is displayed to home cooks while they are reading a recipe from a particular cookbook.
- Each campaign has a campaign number (stored in a sequentially increasing manner starting from 1), advertised grocery item name, advertised grocery item category, the hyperlink of the campaign, and the cashback rate per read (paid to the merchant by the platform).
- Additionally, the domains of the advertised grocery item category and the recipe ingredient are the same.
- Campaigns are not displayed on the cookbook as a whole. In other words, a campaign is associated with a recipe that is displayed once.
- A single campaign can appear in multiple recipes. For each recipe where the merchant decides to display a campaign, a display cost needs to be paid to the platform. However, this cost can vary across recipes.

You are encouraged to **add more detail, if required, to the above data requirements** so that it helps you in the EER modeling phase. *Use red font color for the additions.*

Assumptions

- **User:** We add the city to the list of attributes captured for users to be used for the business goal 6.
- **User:** We add the date of birth as atomic attribute, and full name as composite attribute capturing both first name, last name to the list of attributes captured for users
- **User:** we derive the user's age from date of birth, since it changes instead of considering the user's age as a captured attribute.
- **User:** It is conventional to have unique username and email address. Therefore, we assumed that these two are unique and can be candidate keys.
- **Home Cook:** We made "log" attribute of the Home Cook reads recipes composite and added start time and end time to calculate the duration of the read using how long attribute will be derived.
- **Home Cook:** home cook number of readings will show how many times a home cook read a certain recipe, and it is different from num of reading which show how many time a certain recipe was read by all the home cooks.
- **Chef:** A user can be chef if they have at least one published recipe meaning they should have published at least one cookbook.
- **Merchant:** Each website should have a unique domain, so we are assuming that merchant's website is unique and can be candidate key.
- **Recipe:** We assume the list of 3 main ingredients acts as a composite attribute with exactly three simple sub-components, rather than a multi-valued attribute, allowing it to be mapped directly into the Recipe relation.
- **Read:** I added an interaction type attribute to the relationship read to make it easier to calculate the number of like and dislikes. (Business Goal 2)
- **Home Cook (Reads):** We assume a home cook can read the same recipe multiple times. Therefore, start_time is included alongside the foreign keys as part of the composite primary key for the Reads relation to uniquely identify each reading session and prevent duplicate key violations.

Business Goals

1. For each day of the week, what is the average reading minutes spent by a home cook?
This is helpful for the chefs to understand the home cook activity and decide when to publish a new recipe.
2. List the top 10 chefs who on an average have received the most number of likes across all their cookbook recipes published in 2025.
Such a report helps the home cooks to identify trending chefs. Also, the campaign display cost and cashback rate per read can be adjusted for the recipes published by such top chefs.
3. For each year of birth, list the most popular meal type along with its average recipe cooking time. The popularity of a meal type is quantified based on the total number of reads the related recipes receive.
This report is helpful to gauge the meal type preferences among different age groups. Based on such analysis, new home cooks can be shown relevant content to keep them engaged and hooked to the platform.
4. List the name, cuisine, number of recipes, and the average number of reads per recipe for the cookbooks that have only been read by the students from the 'CSE' department enrolled after July 2024. Order the list in the decreasing order of the average number of reads per recipe.
Such listicles help to identify trending cookbooks among a certain group of people.
5. Who are the student subscribers who have read every recipe from the Indian or Korean cuisine cookbooks that have the phrase 'campfire kitchen' appear somewhere in the description? Find their name, GPA, nationality, and email ID.
Such a report identifies the enthusiasts who are reading specific types of recipes. This set of people can act as a group who have acquired enough relevant knowledge on a certain topic and thus can become a part of menu designing teams for events, or can be contacted for conceptualizing easy fusion recipes.

In the above business goals, the campaign information has not been used. You are required to think and clearly write down **two more interesting business goals which use the captured Merchant and Campaign information, along with other captured data.**

6. List the names, websites, and advertised grocery items of merchants whose company headquarter city is "Arlington" (or matches the student's local city) and who have at least one campaign on the platform. This allows the platform to highlight these specific campaigns to nearby students, helping students find better local prices while driving more traffic and support community businesses.

7. For each merchant, list the total display cost paid to the platform and the total cashback owed, computed as each campaign's cashback rate per read multiplied by the number of reads of every recipe it appears on. Report the net amount for each merchant and order by ascending net amount. Because display cost is negotiated per recipe while cashback accrues per read, a merchant placing a low-cost campaign on a heavily read recipe can collect more than they pay. This report surfaces those cases so the platform can renegotiate the rate or the placement price.